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Transcriptomic Signatures in Autism Spectrum Disorder: Opportunities and Limitations across Brain, Blood, and Saliva

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Abstract

Autism spectrum disorder (ASD) is behaviorally defined yet biologically heterogeneous, which has encouraged interest in transcriptomic readouts that may capture underlying molecular variation. This focused narrative review examines what transcriptomic studies can and cannot currently contribute to ASD research and translation across three principal specimen types: postmortem brain, peripheral blood, and saliva. Evidence from cortical RNA-seq consistently identifies reduced neuronal and synaptic expression programs, altered alternative splicing, and increased immune-glial signatures, while single-nucleus studies refine these observations to specific cell populations. Peripheral blood studies reveal immune-related and cell-composition-sensitive signals, including monocyte- and NK-cell-associated shifts, but their translational value is constrained by small cohorts, preanalytical variability, and limited external validation. A notable non-invasive translational signal comes from salivary small-RNA work, which supports proof of concept for adjunctive biomarker panels rather than stand-alone diagnosis. Across tissues, convergent axes - synaptic dysfunction, immune dysregulation, mitochondrial disturbance, and RNA-processing abnormalities - support biological plausibility, but do not yet establish a clinically deployable transcriptomic test. Platform choice is also consequential: bulk RNA-seq is scalable but insensitive to cellular heterogeneity; single-nucleus RNA-seq provides mechanistic resolution but is not clinically practical; and long-read RNA-seq improves isoform discovery but remains resource-intensive. Current evidence positions transcriptomics primarily as a research tool for mechanistic inference and candidate stratification, with limited but real potential for future adjunctive biomarker development rather than replacement of behavioral diagnosis.

Keywords: RNA sequencing, Biomarker, Mitochondrial dysfunction, and Alternative splicing.

1. Introduction

Autism spectrum disorder (ASD) is a heterogeneous neurodevelopmental condition defined clinically by differences in social communication together with restricted or repetitive patterns of behavior and

sensory atypicalities (American Psychiatric Association, 2013; Lord *et al.*, 2020). Prevalence estimates from U.S. surveillance remain high, underscoring the need for better biological characterization as well as earlier and more reliable identification (Maenner *et al.*,

2023). Although behavioral instruments remain central to diagnosis, the disorder itself is biologically diverse, and this mismatch between a behaviorally defined phenotype and heterogeneous underlying biology has driven continuing interest in molecular biomarkers.

Transcriptomics is attractive in this context because RNA profiles can capture dynamic molecular states more directly than static genomic variants alone. In ASD research, transcriptomic approaches have been used to interrogate postmortem brain tissue, peripheral blood, saliva, and related model systems. These studies have not produced a single diagnostic transcript; instead, they have highlighted coordinated disturbances in synaptic function, immune signaling, mitochondrial pathways, and RNA processing (Voineagu *et al.*, 2011; Parikshak *et al.*, 2016; Gandal *et al.*, 2022). The translational question is therefore not whether one transcript can diagnose ASD, but whether reproducible pathway-level signals from accessible tissues can support stratification or adjunctive testing.

That distinction is important. Several reviews and consortium statements emphasize that ASD biomarkers remain at a research stage and should not be interpreted as replacements for behavioral assessment in individual clinical decision making (Frye *et al.*, 2019; McPartland *et al.*, 2020; Akter *et al.*, 2025). At the same time, transcriptomic studies have substantially advanced the biological understanding of ASD, especially by linking genetic risk, cell-type-specific dysfunction, and downstream molecular programs.

This review reframes the topic as a focused biosciences review rather than a broad clinical diagnostics overview. The aim is to evaluate where transcriptomic evidence in ASD is already biologically informative, where it shows selective translational promise, and where methodological limitations still preclude clinically deployable use across brain, blood, and saliva. A mechanistic overview of these specimen-level opportunities and constraints is shown in **Fig. 1**.

2. Materials and Methods

This article was prepared as a focused narrative review. Formal systematic review procedures, meta-analysis, and PRISMA-based study selection were not applied. Instead, the review selectively synthesizes

peer-reviewed literature cited in the manuscript, with emphasis on human ASD transcriptomic studies using bulk RNA sequencing, single-nucleus RNA sequencing, long-read RNA sequencing, or targeted RNA panels.

Priority was given to studies that addressed at least one of the following: reproducible pathway-level dysregulation, cell-type or isoform resolution, clinically accessible biospecimens, or explicit translational claims regarding diagnostic or stratification utility. Selected preclinical studies were included only when they clarified biological plausibility for transcriptional regulators or pathway-level mechanisms. The review was organized deliberately around specimen type, translational relevance, and interpretability of transcriptomic findings rather than exhaustive coverage of all published ASD RNA studies.

3. Results and Discussion

Biological rationale for transcriptomic biomarkers in ASD

ASD is genetically complex, developmentally dynamic, and phenotypically heterogeneous. Under such conditions, transcriptomics is informative because it can integrate upstream influences from common and rare variants, developmental timing, cellular context, and environmental modulation. The most consistent message from the field is not the discovery of a single ASD-specific RNA marker, but the recurrence of coordinated molecular programs: neuronal and synaptic downregulation, immune and glial activation, altered splicing, and mitochondrial or energetic disturbance (Voineagu *et al.*, 2011; Parikshak *et al.*, 2016; Schwede *et al.*, 2018; Gandal *et al.*, 2022). The translational value of these patterns lies in whether they remain measurable, interpretable, and reproducible in clinically accessible tissues.

Brain transcriptomic evidence: mechanistic depth with limited clinical accessibility

Postmortem brain tissue provides the strongest mechanistic transcriptomic signal in ASD. Early cortex-based work identified convergent molecular pathology characterized by reduced synaptic gene expression together with increased immune and glial signatures (Voineagu *et al.*, 2011; Mondol *et al.*, 2018). Subsequent RNA-seq studies extended this

picture by showing genome-wide changes in long non-coding RNAs, splicing, and regional expression programs, reinforcing the view that altered RNA processing is part of ASD biology rather than a secondary effect (Parikshak *et al.*, 2016). More recent

cortex-wide analysis demonstrated broad transcriptomic dysregulation across the cerebral cortex, indicating that these changes are not confined to one narrowly sampled region (Gandal *et al.*, 2022).

Representative Transcriptomic evidence across major ASD biospecimen types

Table 1: Representative Transcriptomic evidence across major ASD biospecimen types.

Study	Biospecimen	Design / platform	Main finding	Translational note
Voineagu <i>et al.</i> , 2011	Postmortem cortex	Transcriptomic analysis of autistic brain	Convergent reduction of synaptic programs with increased immune / glial signatures	Strong mechanistic evidence; not clinically accessible
Parikshak <i>et al.</i> , 2016	Postmortem cortex	RNA-seq	Genome-wide changes in lncRNAs, splicing, and regional expression	Supports RNA-processing abnormalities in ASD
Velmeshev <i>et al.</i> , 2019	Postmortem cortex nuclei	Single-nucleus RNA-seq	Cell-type-specific changes in excitatory neurons and microglia	High mechanistic resolution; not practical for diagnosis
Gandal <i>et al.</i> , 2022	Cerebral cortex	Cortex-wide RNA-seq	Broad transcriptomic dysregulation across cortical regions	Reinforces convergence of brain-level pathology
Filosi <i>et al.</i> , 2020	Peripheral blood	RNA-seq plus deconvolution	ASD signal partly reflects altered immune-cell composition, including reduced NK-cell signatures	Highlights need for composition-aware analysis
Tomaiuolo <i>et al.</i> , 2023	Peripheral blood	Discordant siblings; RNA-seq / WGCNA	Immune-related DE genes and reduced coexpression module with modest discrimination	Useful biologically, limited diagnostic performance
Li <i>et al.</i> , 2023	Peripheral blood	Bioinformatic deconvolution plus clinical validation	Myeloid / monocyte shifts with moderate ROC performance for selected hub genes	Supports peripheral immune involvement; not sufficient alone
Voinsky <i>et al.</i> , 2022	Whole blood	RNA-seq and RT-qPCR follow-up	Altered BATF2, LY6E, ISG15, and MT2A expression	Candidate panel components requiring larger validation
Voinsky <i>et al.</i> , 2023	Whole blood	Machine-learning targeted panel	Approx. 82% accuracy in internal validation settings	Proof of concept; external transportability unresolved
Hicks <i>et al.</i> , 2018	Saliva	Small-RNA and microbial panel	Independent-test AUROC ~0.88 with 82% sensitivity and 88% specificity	Most promising non-invasive adjunctive signal to date

Single-nucleus RNA-seq added an important layer of resolution by showing that transcriptomic abnormalities are not uniformly distributed across all cells. Instead, major shifts were concentrated in specific neuronal and glial populations, including upper-layer excitatory neurons and microglia (Velmeshev *et al.*, 2019). This matters because bulk tissue averages can obscure whether a signal reflects altered cell composition, altered per-cell expression, or both. Brain data therefore provide high biological credibility for transcriptomic dysregulation in ASD, but they are intrinsically limited as screening or diagnostic substrates because postmortem tissue is inaccessible

during routine pediatric assessment and most available cohorts do not reflect the earliest developmental window.

Brain studies also point to pathway-rich mechanistic nodes rather than clinical biomarkers per se. Synaptic genes and scaffolding systems such as SHANK3, as well as isoform-level disturbances involving genes such as ANK2, illustrate how transcriptomics can reveal regulatory and structural complexity not captured by gene-level summaries alone (Gandal *et al.*, 2018; Lu *et al.*, 2024; Wang *et al.*, 2014). Yet these findings should not be overinterpreted: they

improve mechanistic understanding, but they do not by themselves define a deployable test.

Peripheral blood transcriptomics: accessible but composition-sensitive

Peripheral blood is attractive because it is clinically obtainable, amenable to repeat sampling, and compatible with standard molecular workflows. However,

blood is also one of the most composition-sensitive transcriptomic matrices. Changes in leukocyte proportions can strongly influence bulk RNA profiles, which mean that an observed ASD-associated signature may partly reflect cell mixture rather than cell-intrinsic dysregulation. This issue is central, not secondary, to interpretation.

Biospecimen-level opportunities and limitations

Table 2: Opportunities and limitations of major biospecimens in ASD transcriptomics.

Biospecimen	Main opportunity	Main limitation	Most realistic role
Brain tissue	Highest mechanistic relevance; strong pathway and cell-type signals	Unavailable for routine diagnosis; often from older postmortem cohorts	Mechanistic discovery and target identification
Peripheral blood	Clinically obtainable, repeat sampling, compatible with standard molecular workflows	Strongly affected by leukocyte composition, batch effects, and preanalytics	Candidate stratification or adjunctive testing after stronger validation
Saliva	Non-invasive and pediatric-friendly; practical for repeated collection	Evidence base still narrow; transportability and generalizability not established	Most feasible future adjunctive screening-support matrix

Several blood studies nonetheless provide important signals. Filosi *et al.* (2020) reported transcriptome differences in discordant sibling pairs and showed that part of the ASD signal was explained by altered peripheral immune cell composition, particularly reduced NK-cell-associated signatures. Tomaiuolo *et al.* (2023), in sex- and age-matched discordant siblings, identified differential expression of immune-related genes together with a reduced coexpression module whose eigengene produced only modest discriminatory performance. Li *et al.* (2023) combined bioinformatic deconvolution with clinical validation and described increased monocytes and other myeloid shifts in children with ASD, while Horiuchi *et al.* (2021) reported aberrant innate and adaptive immune signaling in adults. Collectively, these studies support the existence of peripheral immune dysregulation, but they also show why blood signatures can be unstable across cohorts, ages, and analytical pipelines.

Targeted blood-panel work illustrates both the opportunity and the limitation of translation. Voinsky *et al.* (2022) identified altered whole-blood expression of BATF2, LY6E, ISG15, and MT2A, and later developed a machine-learning blood RNA signature that achieved approximately 82% accuracy under

internal validation settings (Voinsky *et al.*, 2023). These data are encouraging as proof of concept, but they remain insufficient for clinical deployment. Sample sizes were modest, classifier performance was not established in large external multicenter cohorts, and overestimation risk remains whenever feature selection and model optimization are closely tied to limited datasets (Varma and Simon, 2006). Blood transcriptomics therefore remains best viewed as a candidate stratification layer rather than a mature diagnostic platform.

Salivary RNA evidence: the most practical pediatric matrix

Among non-invasive matrices, saliva is especially attractive for pediatric research because sampling is simple, repeatable, and far less burdensome than venipuncture. A notable translational signal in the present literature comes from the salivary RNA work of Hicks *et al.* (2018), who reported a multicenter panel comprising small RNAs together with microbial taxa and obtained an AUROC of approximately 0.88 in an independent test set, with sensitivity near 82% and specificity near 88%. Importantly, the authors did not frame the assay as a replacement for behavioral diagnosis, but as an adjunctive tool after screening or

as an objective support signal. That framing is appropriate. Salivary RNA signatures offer real translational promise, but the evidence remains bounded by several limitations. The study was not a prospective population-level screening study, geographic and dietary effects may influence microbial components,

and broader transportability across laboratories and populations remains unproven. Saliva should therefore be considered the most practical current matrix for future adjunctive biomarker development, not proof that a stand-alone molecular diagnostic test for ASD already exists.

Platform comparison

Table 3: Comparison of major transcriptomic platforms relevant to ASD biomarker research.

Platform	Primary strength	Primary weakness	Best current use
Bulk RNA-seq	Scalable discovery of differential expression and pathways	No cell-type resolution in mixed tissues	Primary discovery in brain and blood
Single-nucleus RNA-seq	Cell-type-specific attribution of ASD-related signals	High cost and technical complexity	Mechanistic dissection of brain pathology
Long-read RNA-seq	Improved isoform and transcript-structure resolution	Lower throughput and quantification challenges	Follow-up of splicing and isoform questions
Targeted RNA panels	Most clinically practical endpoint	Dependent on prior discovery and vulnerable to overfitting	Prospective validation of locked biomarker sets

Cross-tissue convergence and what it really means

One of the strongest arguments for continued transcriptomic research in ASD is cross-study convergence at the level of pathways. Brain tissue repeatedly supports reduced synaptic and neuronal programs, immune activation, and altered splicing. Blood studies repeatedly implicate immune-cell-related signals, especially myeloid and NK-cell-associated shifts. Mitochondrial and energy-related pathways also recur, and in brain tissue they correlate with synaptic downregulation rather than appearing as isolated abnormalities (Schwede *et al.*, 2018). This convergence supports biological plausibility.

However, convergence does not mean equivalence. A blood immune signature is not a direct surrogate for cortical dysfunction, and a salivary small-RNA classifier is not a window into the whole brain. The more defensible interpretation is that different tissues provide different levels of biological access. Brain tissue offers mechanistic depth; blood offers systemic accessibility but heavy compositional confounding; saliva offers pediatric feasibility with a narrower evidence base.

Translation will depend on whether accessible matrices can capture a clinically useful subset of the relevant biology in a reproducible way.

Platform-specific opportunities and constraints

Platform choice materially affects what can be learned from ASD transcriptomics. Bulk RNA-seq remains the workhorse for discovery because it is scalable, relatively mature analytically, and well suited to differential expression and pathway enrichment studies (Wang *et al.*, 2009; Conesa *et al.*, 2016). Its main weakness is that it averages across mixed cell populations. Single-nucleus RNA-seq addresses that limitation and has been critical for cell-type attribution in ASD brain tissue, but it is technically demanding, costly, and poorly suited to routine clinical workflows (Velmeshev *et al.*, 2019). Long-read RNA-seq improves full-length isoform identification and is relevant to a disorder in which splicing and transcript architecture matter, yet throughput and quantification challenges still limit its use as a front-line translational platform (Pardo-Palacios *et al.*, 2024; Ament *et al.*, 2025). Targeted panels, including qPCR-based or small-RNA assays, are the most plausible clinical endpoint, but only after robust discovery and external validation.

Translational limitations: why the field is not clinically ready yet

Current transcriptomic evidence is promising but not clinically definitive for several reasons. First, many studies are still small, case-control oriented, and

vulnerable to cohort-specific effects. Second, pre-analytical variation - sampling tubes, stabilization time, storage, extraction method, and library preparation - can introduce substantial technical variability. Third, batch effects and composition effects remain major threats to portability and should be modeled explicitly rather than discussed post hoc (Sprang *et al.*, 2022). Fourth, performance estimates for classifier studies are often derived from internal cross-validation and may overstate real-world accuracy when not supported by locked models and independent external test cohorts (Varma and Simon, 2006).

These limitations explain why systematic reviews and consortium initiatives still position ASD biomarkers, including transcriptomic candidates, as adjunctive and investigational rather than clinically qualified tools (Frye *et al.*, 2019; McPartland *et al.*, 2020). For a transcriptomic test to become persuasive, future studies will need developmentally relevant cohorts, harmonized sampling procedures, explicit control of cell composition, predefined analytical pipelines, and true external validation across sites. Without that level of rigor, the field risks confusing biological signal with diagnostic readiness.

Mechanistic overview of transcriptomic opportunities and limitations in ASD

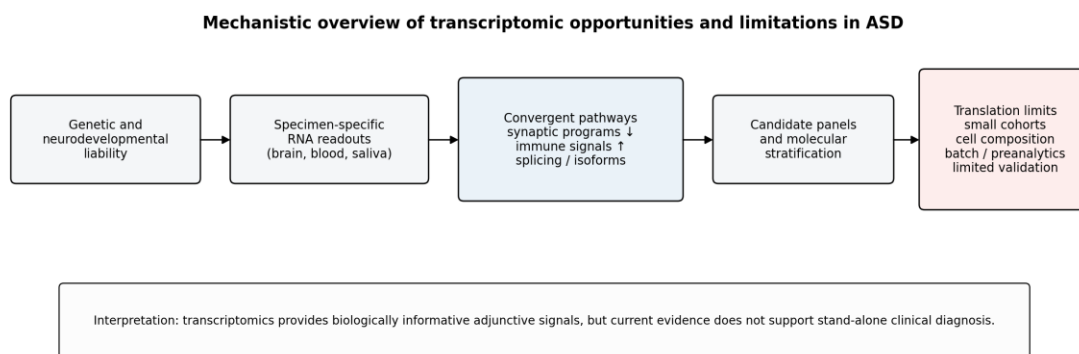


Fig. 1: Mechanistic overview of transcriptomic opportunities and limitations in ASD across brain, blood, and saliva. Convergent transcriptomic signals support biological plausibility, whereas small cohorts, cell-composition effects, preanalytical variability, and limited external validation constrain present-day clinical use.

4. Conclusion

Transcriptomic studies have already made a substantial contribution to ASD biology, but their impact on clinical practice remains limited. The strongest evidence comes from postmortem brain tissue, where reproducible synaptic, immune, splicing, and mitochondrial abnormalities support the existence of convergent molecular pathology. Peripheral blood and saliva extend this work into clinically accessible matrices and therefore define the principal translational direction for future biomarker research, rather than an already established diagnostic pathway. At present, however, those peripheral signals remain too sensitive to cohort structure, cell composition, technical variation, and limited external validation to justify stand-alone clinical use. The most realistic near-term role for transcriptomics is as a research tool for mechanistic stratification and, potentially, as an

adjunctive layer alongside behavioral assessment rather than a substitute for it. Progress toward a clinically credible RNA-based assay will require pathway-informed biomarker selection, standardized preanalytics, locked analytical models, and multicenter prospective validation in pediatric populations. The translational opportunity is genuine, but clinically credible implementation will require a substantially stronger evidence base than is currently available.

5. Ethical Clearance

This article is a narrative review and does not report new experiments involving human participants, animals, or identifiable personal data. Ethical approval was therefore not required.

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7. Conflicts of Interest

The author declares no conflicts of interest.

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